

35 TALLEST Space Rockets Ever Launched!

Bold text to be shown on the screen

HOOK:

Some rockets flew once. Others made history. But these? They shattered height records.

And #1? It might just replace airplanes forever.

INTRO

Watching **Starship's** Super Heavy booster thunder into the sky at **123 Meter (403.5 Foot)** is like witnessing tomorrow's space age in action, today.

add a clip with starship footage

But how did we go from that stainless-steel giant to the first "tower of fire" back in 1967? That honor goes to **Saturn V**, which rose **110.6 Meter (362.8 Foot)** off Pad 39A and carried us to the Moon. Since then, engineers have raced to push the limits of height, strength, and ambition.

Today, we're counting down the **35 tallest rockets** ever built, and we kick off with the giants that redefine what "reach for the stars" really means, then work our way down through history's most daring launch vehicles.

We'll start with the sky-scraping behemoths and then peel back the layers of history's most daring launch vehicles!

Buckle up! Our climb begins now.

THE SUPER HEAVIES (98 m and up)

At **98 meters (321 Foot)**, **NASA's SLS Block 1** entered the stage, powered by four **RS-25 shuttle engines** and twin solid rocket boosters. It made history by sending the uncrewed **Orion capsule** around the Moon in **2022**, the first step of **NASA's Artemis program**, aiming to return humans to the lunar surface. Standing nearly as tall as the **Statue of Liberty**, the SLS Block 1 is a towering symbol of America's return to deep space exploration.

Then there's the one-hit wonder: **Ares I-X**. Reaching exactly **100 meter (328 foot)** during its **2009 test flight**, it briefly held the title of the **tallest rocket of the 21st century** before being shelved. A glimpse of what could have been.

But the true king of them all? **Saturn V**. Standing at an incredible **110.6 meter (362.8 foot)**, this rocket carried **24 astronauts** to the Moon between **1968 and 1972**, including the historic **Apollo 11** mission. Its legacy is unmatched, setting the standard for what a rocket can achieve in terms of **power, precision, and human ambition**.

The **Soviet Union**, not to be outdone, reached even higher with the **N1**, standing at **114 meter (374 foot)**. Powered by a staggering **30 NK-15 engines**, it was designed to match the **Saturn V** but suffered four spectacular failures. Still, it symbolizes the **bold ambition** of the Cold War era.

And now, the towering behemoth (**image of Behemoth**): **Starship**. With its **Super Heavy booster stacked**, it stands at an eye-watering **123 meter (403.5 foot)**, holding the title of the **tallest rocket ever built**. Already having completed several **test flights** from **Boca Chica**, it's gearing up for its **first-ever booster catch attempt** in **late 2025**. **Starship** isn't just tall, it's **rewriting the future of space travel**, pushing the boundaries of what's possible in the space industry.

UPPER SIXTIES TO LOW SEVENTIES (66 m – 71 m)

Now, we step into the shadow of true giants... rockets that not only shaped history but lit up the skies with raw power and bold promises. These weren't just launch vehicles; they were **symbols of ambition, engineering excellence, and national pride**.

First, at **66.4 meter (217.8 foot)**, the **Delta IV Medium+** stood tall as part of the Delta IV family. Outfitted with four powerful **GEM-60 solid boosters**, it was a quiet workhorse that didn't make headlines but consistently delivered vital national security payloads to orbit. It served reliably until its retirement in **2018**, playing a behind-the-scenes role in some of the **U.S. military's most sensitive space missions**. It wasn't the most glamorous rocket, but when precision and trust were needed, it delivered! Standing just a few feet shorter than the **Leaning Tower of Pisa**, it packed power into a compact frame.

Next, at **68.3 meter (224 Foot)**, comes **NASA's Saturn IB**, complete with its unmistakable **escape tower**. This rocket played a key role in the **Apollo program** and **Skylab missions**, especially **Apollo 7**, which marked the United States' return to crewed spaceflight after the tragic **Apollo 1** fire. Though it lived in the shadow of the legendary **Saturn V**, the **Saturn IB** proved its worth time and again in the late '60s and early '70s. It was dependable, well-designed, and crucial for getting astronauts safely into Earth orbit.

Then we arrive at the iconic **Falcon Heavy**, towering at **70 meter (229.6 foot)**. Built by strapping together three **Falcon 9** boosters, this rocket became a symbol of the **new space age**. In **2018**, it captured the world's attention by launching **Elon Musk's personal Tesla Roadster**,

complete with a mannequin in a spacesuit, into orbit around the Sun. Beyond the theatrics, **Falcon Heavy** changed the economics of heavy-lift launch. Its **reusable core stages** and **record-setting performance** offered a glimpse into a future where rockets could do more for less, again and again.

Finally, at **70.7 meter (232 foot)**, we meet the mighty **Delta IV Heavy**. Its three **Common Booster Cores** gave it immense thrust, sending some of the **heaviest and most secretive government payloads** into orbit. Its unmistakable **hydrogen-fueled orange flame** and thunderous liftoff were a spectacle to witness. On **April 9, 2024**, it launched for the final time, closing the chapter on one of the most **powerful expendable rockets** America ever built. A true heavyweight in both **legacy** and **capability**.

[LOW- AND MID-FIFTIES \(53 m – 56 m\)](#)

Crossing the fifty-meter mark, we meet **Europe's now-retired Ariane 5 ECA**, standing tall at **53 meter (173.9 foot)**. Its long career ended on a high note...the flawless launch of the **James Webb Space Telescope in 2021**, one of the most ambitious science missions ever sent to space. Matching it in height is **Japan's H-IIA 202**, still going strong with a near-perfect success rate, lifting scientific payloads and satellites with remarkable reliability. Both rockets stood about as tall as the **Hollywood Sign's letters**; giants in performance, if not in size.

Just a bit taller at **53.7 meter (176.2 foot)**, **China's Long March 5B** made headlines not only for launching major modules of the Tiangong space station, like **Tianhe and Wentian**, but also for the controversy around its uncontrolled re-entries. Standing even higher at **55 meter (180.4 foot)** is **Saturn I**, NASA's early testbed in the 1960s. Though often overshadowed by **Saturn V**, this rocket quietly laid the groundwork, proving that a cluster of engines could safely power Apollo-era missions.

At **55.4 meter (181.7 foot)**, Russia's **Angara A5** represents a modern shift in its space program, offering a cleaner and more modular alternative to older launch vehicles, flying from the newer Vostochny Cosmodrome. Then there's the full **Space Shuttle** stack at **56.1 meter (184 foot)**, an icon of human spaceflight. With **135 missions**, it brought satellites, science labs, and astronauts to space, though not without **tragedy and triumph along the way**.

Nearly matching it is **China's Long March 3B** at **56.3 meter (184.7 foot)**, a reliable workhorse for geostationary satellite launches. Finally, **Japan's H-IIB** rounds out this group at **55 meter (180.4 foot)**, remembered for carrying large HTV cargo ships packed with vital equipment for the ISS; payloads few other vehicles could handle.

THE FORTIES (42 m – 49 m)

We begin with the **Titan IIIC**... a **42 meter (137.8 foot)** pillar of power that first roared to life in **1965**. It sent early GPS satellites and classified payloads into orbit, pushing the boundaries of space exploration at the time. But did you know it paved the way for future military and civilian missions alike? **Antares 230+** comes in just half a meter taller, best known for sending Northrop Grumman's Cygnus cargo ships up to the ISS. Wondering how much it could haul? Plenty... enough to keep the station stocked and running for years!

Next up is India's **LVM3**, standing tall at **43.5 meter (142.7 foot)**. It's **one of the country's most powerful rockets** to date, built for big missions and heavy payloads. Just a little taller is the PSLV-XL at 44 meters, known as the 'workhorse' of Indian space missions. With over **400 satellites launched**, how does it keep going strong? Simple: it's **reliable, proven, and built to last**.

Close behind is **Russia's Soyuz-2.1b**, standing at **46.9 meter (153.9 foot)**. It's a dependable workhorse in Russia's fleet, regularly used to launch OneWeb satellites and deliver cargo to the ISS. Known for its consistency and solid performance, it's been a key player in recent years. And just a bit taller is **South Korea's KSLV-II Nuri** at **47.2 meter (154.8 foot)**; a fully domestically developed rocket that's steadily proving itself. After a third successful launch in November 2024, can it continue to make waves in space exploration? Only time will tell...

OUTRO

From the **42-meter (137.8 Foot) Titan II** to the **123-meter (403.5 foot) Starship**, rockets keep getting taller... and so do our dreams. But stature isn't glory, it's the payloads delivered and legends made.

What's your pick: Saturn V's thunder, Vulcan's fresh fire, or Starship's sky-scraping beast? **Drop your favourite below, smash like, and hit subscribe because the sky isn't the limit... it's just the launchpad.**

References:

<https://www.space.com/12944-worlds-tallest-rockets-comparison.html/>

<https://www.businessinsider.com/watch-spacex-starship-launch-oft-2-elon-musk-rocket-2023-1>

<https://apollo11space.com/worlds-biggest-rockets/>

<https://www.visualcapitalist.com/comparing-the-size-of-the-worlds-rockets-past-and-present>

<https://www.sciencetimes.com/articles/39330/20220812/side-comparison-nasa-s-sls-saturn-v-cost-height-weight.html/>

<https://time.com/6252046/spacex-starship-rocket/>

<https://www.reuters.com/science/spacex-gets-faa-approval-ramp-up-starship-launches-texas>

<https://www.nasa.gov/wp-content/uploads/2023/07/0080-sls-fact-sheet-jun2023-508-1.pdf>

<https://www.space.com/ula-vulcan-centaur-first-launch-peregrine-celestis-moon-mission>

https://en.wikipedia.org/wiki/Super_heavy-lift_launch_vehicle